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Swapnil Bhole

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Professional Summary

ECE Master's student at the University of Michigan with 6 years of industry experience in embedded firmware, SoC-level system integration, and hardware validation. Proven track record writing low-level firmware in C on ARM platforms, executing HW bring-up on FPGA and SoC hardware, and building test automation in Python. Hands-on with RTOS concepts, peripheral protocols (SPI, I2C, UART, JTAG), 3GPP protocol compliance testing, and pre/post-silicon validation workflows.

Education

University of Michigan, Ann Arbor **Aug 2024 — May 2026**

- Master of Science in Electrical and Computer Engineering GPA: 3.7/4.0
- Coursework: Computer Architecture, Parallel Computer Architecture, Applied GPU Programming, Embedded Systems, Digital Communication

VNIT Nagpur, India **Jul 2014 — Aug 2018**

- Bachelor of Engineering in Electronics and Communication Engineering GPA: 8.9/10.0

Selected Projects

2-Way Superscalar Out-of-Order RISC-V Processor **Sep-Nov 2024**

- Designed and verified a full out-of-order RISC-V CPU in SystemVerilog with R10K-style register renaming, branch prediction, and separate I/D caches; validated using Synopsys VCS, achieving an average CPI of 3.96; version-controlled across the team using Git

CUDA Acceleration of CNN Inference **Oct-Nov 2025**

- Optimized CNN convolution layers using custom CUDA kernels; reduced inference runtime from 13+ seconds to 34 ms (380x speedup) through improved memory access patterns and parallel thread execution on GPU hardware

ESP32-Based IoT Coffee Station **Oct-Nov 2025**

- Built a bare-metal embedded application on ESP32 integrating an SPI-driven LCD display and an I2C capacitive touch controller; implemented peripheral drivers in C for register-level hardware interaction, real-time sensor polling, and display rendering

ML-Based Non-Coherent Wireless Modulation Scheme **Jan-Apr 2025**

- Developed a machine learning joint coded modulation scheme for non-coherent detection; outperformed conventional LoRa in link reliability, demonstrating applied knowledge of PHY-layer signal processing and wireless systems

Experience

Centre for Development of Telematics (C-DOT), Bengaluru **Aug 2018 — Jul 2024**

India's national telecom R&D lab under the Ministry of Communications

Senior Engineer (Level EII) **2021 — Jul 2024**

- Developed 5G Low-PHY design on Zynq-7000 SoC (Kintex-7 FPGA + dual-core ARM Cortex-A9) running PetaLinux; delivered a production-deployable embedded application integrating firmware, OS, and RF hardware layers
- Implemented 5G NR PHY-layer signal processing modules using MATLAB System Generator and Vivado IPs; validated end-to-end against Keysight 5G NR simulator via JTAG and instrument interfaces
- Designed a duo-binary turbo decoder for DVB-RCS in MATLAB HDL Coder

Engineer (Level EI) **Aug 2018 — 2021**

- Developed and executed 3GPP protocol compliance test suites (Releases 9–12) covering S1, X2, and Uu interface protocol stacks for LTE eNodeB; results directly secured a major government telecom tender
- Performed eNodeB performance and interoperability testing for throughput, load, and stability using Amarisoft, TM500, and Keysight UXM; validated VoLTE, NB-IoT, and 2x2 MIMO across FDD and TDD configurations

Publications

- ATRNN: Using Seq2Seq Approach for Decoding Polar Codes, *COMSNETS*, 2020
- DNNStream: Deep-learning based Content Adaptive Real-time Streaming, *SPCOM*, 2020
- Deep Learning based Object Detection Framework for the Visually-Impaired, *ICCMC*, 2020
- Compact Planar Antenna for TV White Space Applications, *SCEECS*, 2018